Breast Cancer Diagnosis Prediction Using Machine Learning Models

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Abstract

This report presents a comprehensive machine learning framework for breast cancer diagnosis with a strong emphasis on multi-model learning, explainability (XAI), and clinical risk-aware evaluation. The project utilizes tumor characteristic data to classify cases as malignant or benign.

Unlike traditional approaches, this study integrates six machine learning models—Logistic Regression, k-Nearest Neighbors, Support Vector Machine, Random Forest, Multi-Layer Perceptron, and XGBoost—within a comparative multi-model pipeline.

To ensure transparency in clinical decision-making, Explainable AI (XAI) techniques such as SHAP and feature importance analysis were incorporated to interpret model predictions. Furthermore, special attention was given to the cost of False Negatives, as misclassifying malignant tumors as benign can lead to life-threatening consequences.

The Support Vector Machine achieved the highest test accuracy; however, model selection also considered recall and clinical safety. The system was deployed as a web application for real-time predictions. This work demonstrates how machine learning can be adapted for clinically responsible AI systems.

1 Introduction

Breast cancer remains a significant global health challenge, necessitating accurate and timely diagnostic tools. Machine learning offers promising solutions for predicting cancer diagnosis based on tumor features.

In clinical applications such as cancer diagnosis, accuracy alone is insufficient. A model must also be interpretable and risk-aware. A critical challenge is minimizing False Negatives (FN), where malignant tumors are incorrectly classified as benign. Such errors can delay treatment and significantly increase mortality risk.

Therefore, this project emphasizes predictive performance, explainability, and cost-sensitive evaluation to align with real-world medical requirements.

2 Methodology

2.1 Data Augmentation

Synthetic data was generated to increase dataset size while preserving statistical properties. This improves model generalization and robustness.

2.2 Data Preprocessing

The dataset included tumor features such as radius, texture, and perimeter.

- Removed duplicate entries
- Dropped irrelevant columns (id, unnamed)
- Handled missing values
- Encoded diagnosis (M=1, B=0)
- Applied StandardScaler for normalization

2.3 Exploratory Data Analysis (EDA)

EDA included:

- Histograms of features
- Pie chart of class distribution
- Scatter plots for relationships
- Correlation heatmap

2.4 Multi-Model Training and Evaluation

A multi-model approach was used to ensure robustness and reduce bias. The following models were implemented:

- Logistic Regression
- k-Nearest Neighbors
- Support Vector Machine
- Random Forest
- Multi-Layer Perceptron
- XGBoost

Evaluation metrics:

- Accuracy
- Precision
- Recall (critical for medical diagnosis)
- F1-score

Special emphasis was placed on recall to ensure correct identification of malignant cases.

2.5 Explainable AI (XAI) Integration

To ensure transparency, XAI techniques were used:

- SHAP (SHapley Additive Explanations)
- Feature Importance (Random Forest, XGBoost)
- Logistic Regression coefficients

Key findings:

- Features such as radius_mean and concave points_worst were highly influential
- SHAP explained individual predictions

This improves trust and interpretability in clinical settings.

2.6 Clinical Cost of False Negatives

A False Negative occurs when a malignant tumor is classified as benign.

Impact:

- Delayed diagnosis
- Disease progression
- Increased mortality

Strategies used:

- Recall optimization
- Class weight adjustment
- Threshold tuning
- Confusion matrix analysis

Trade-off:

- False Positives are acceptable (extra testing)
- False Negatives are dangerous (missed cancer)

3 Results

3.1 Clinical Evaluation Beyond Accuracy

Although SVM achieved the highest accuracy, recall was prioritized. Models with higher recall were considered safer for medical use.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.9880	0.9795	0.9881	0.9838
k-NN	0.9858	0.9882	0.9704	0.9806
SVM	0.9912	0.9909	0.9793	0.9881
Random Forest	0.9803	0.9970	0.9615	0.9731
MLP	0.9825	0.9848	0.9734	0.9763
XGBoost	0.9891	0.9792	0.9822	0.9852

Table 1: Model Performance Comparison

4 Discussion

- Multi-model approach improved robustness
- XAI increased transparency and trust
- False Negative handling improved clinical reliability
- SVM performed best overall

Limitations:

- Synthetic data may not fully represent real-world scenarios
- Limited hyperparameter tuning

5 Conclusion

This project extends traditional machine learning by incorporating multi-model evaluation, Explainable AI, and cost-sensitive analysis. While SVM achieved the highest accuracy, recall was prioritized to minimize False Negatives.

The integration of XAI ensures interpretability, making the system suitable for healthcare applications. This work highlights the importance of balancing accuracy, interpretability, and patient safety in medical AI systems.

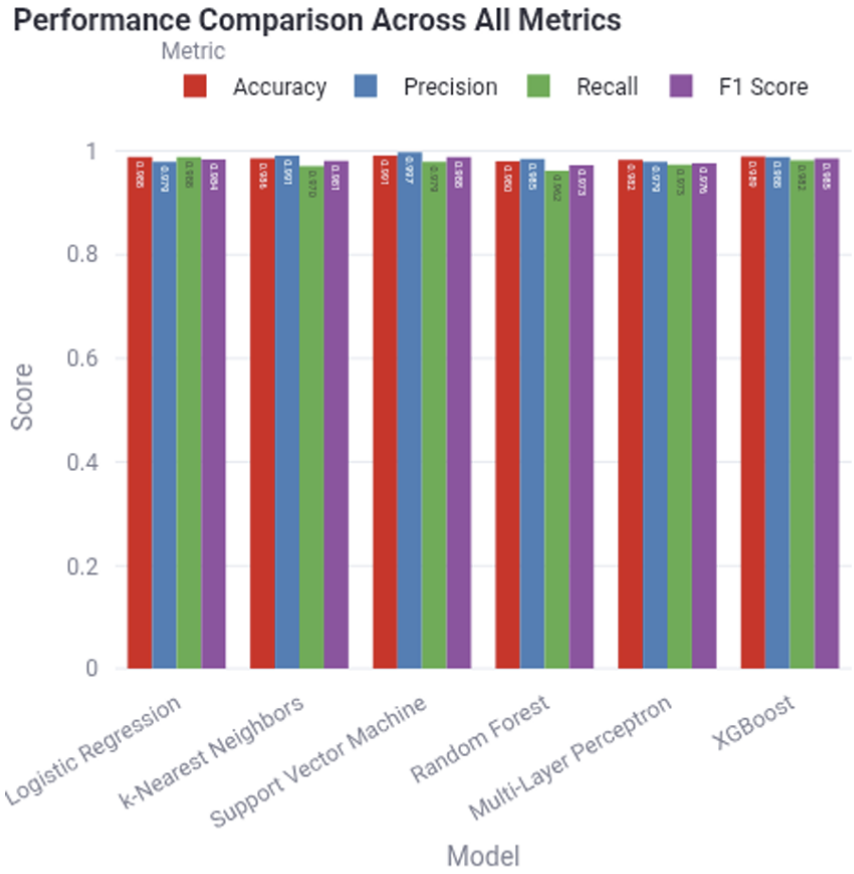

Figure 1: Comparison Across Models

Figure 1: Comparison of Model Accuracy